

A Physics-in-the-Loop Digital Twin with a Dual-Loop Control Framework for Safe, Energy-Optimal Data-Center Cooling

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Abstract. Cooling typically accounts for 30–40% of a data center’s non-IT electricity, yet the inlet-air temperature of every server is a hard safety constraint that operators are unwilling to risk. We present a *digital twin dual-loop control framework* that recommends weekly cooling setpoints for a tropical, water-cooled hyperscale hall while guaranteeing that no server inlet ever exceeds 26°C . An *inner planning loop* treats a full-building EnergyPlus 9.5 model — coupled in the loop through Docker and BCVTB — as the physical oracle: a coarse-to-fine beam search proposes three global setpoints (supply-air temperature, CRAH airflow, and chilled-water supply temperature), each candidate is scored by a complete week-long physics simulation, and a three-layer safety gate rejects anything it cannot *prove* robust. An *outer deployment loop* pre-validates the chosen plan, expands it to 45 per-actuator commands, deploys it in shadow mode, measures the realized week, and learns per-KPI bias and two uncertainties that feed back into both the search margin and a worst-case robust ensemble; a slower physics-recalibration step corrects the plant’s fan-efficiency parameters from persistent energy residuals. Over six consecutive deployed weeks the twin’s weekly-energy prediction error fell from 1.33% to 0.10%, energy savings against the as-operated baseline ramped from 0% to 3.6% — released only as fast as measured confidence allowed — and *zero* safety violations occurred (worst realized inlet 25.60°C). We give the full physical and statistical formulation, including the energy and recirculation models, the conditional-value-at-risk robustness gate, and the empirical-Bayes uncertainty that sizes it, and we illustrate the operator-facing web application that drives the loop.

Keywords: digital twin; data-center cooling; EnergyPlus co-simulation; safe optimization; beam search; robust feasibility; conditional value-at-risk; calibration; energy efficiency.

1 Introduction

Data centers consumed an estimated 1–2% of global electricity in the mid-2020s, and within a facility the cooling system is the dominant non-computing load, commonly 30–40% of overhead power [1, 2, 3]. The efficiency of a center is summarized by its *Power Usage Effectiveness* (PUE), the ratio of total facility power to IT power; every reduction in cooling power moves PUE toward its ideal value of 1. At the same time, cooling is a *safety-critical* function: each server draws air at an inlet whose temperature must stay within the envelope recommended by ASHRAE Technical Committee 9.9 [4]. Exceeding the inlet limit risks thermal throttling, accelerated hardware failure, and — in the tail — thermal shutdown of a production hall. Operators therefore run cooling conservatively, leaving large but *unrealized* efficiency gains on the table because no recommendation can be trusted that has not been shown to be safe.

This tension — aggressive energy optimization versus an inviolable thermal constraint under model and weather uncertainty — is precisely what a *digital twin* [5] is suited to resolve. A high-fidelity physics model of the plant can predict, before anything is actuated, what a proposed setpoint will do to both energy and to the hottest server inlet in the hall. Prior data-center control work has explored machine-learning surrogates [9], model-predictive control over learned dynamics [10], and reinforcement-learning testbeds built on EnergyPlus itself [11]. These approaches optimize well on average but treat the safety constraint softly, or rely on a learned model whose extrapolation behaviour near the constraint boundary is hard to certify. Our design takes the opposite stance: the constraint is hard and is checked by the same first-principles building-physics engine, EnergyPlus 9.5 [6, 7], that the industry uses for design, coupled *in the loop* through the Building Controls Virtual Test Bed (BCVTB) [8].

We make the following contributions.

1. **A dual-loop framework** that cleanly separates a fast *inner planning loop* (forecast → physics oracle → search → safety gate) from a slow *outer deployment loop* (pre-validation → expert approval → shadow deploy → measure → learn), with EnergyPlus as the in-the-loop oracle on *both* sides.
2. **A complete physical and statistical formulation** of the controller: the hall-scoped energy objective, the recirculation mixing model that connects supply-air setpoints to server inlets, a coarse-to-fine beam search over three global setpoints broadcast to 45 actuators, and a three-layer safety architecture whose robustness is sized by an empirical-Bayes posterior uncertainty.
3. **A measured, closed-loop demonstration** over six consecutive weekly cycles, in which prediction error, energy savings, and the safety record are reported from the live system rather than asserted, together with the operator-facing web application that makes the loop usable.

The remainder of the paper is organized as follows. Section 2 describes the physical plant and the dual-loop architecture. Section 3 gives the physical and mathematical formulation of the oracle, the control parametrization, the objective, the forecaster, and the recirculation model. Section 4 details the beam-search optimizer and the experiment establishing that the optimization is physically well-posed. Section 5 presents the three-net robust safety gate. Section 6 describes calibration and physics recalibration — how the twin learns. Section 7 presents the web application and the operator workflow, with screenshots. Section 8 reports the six-week deployment results. Section 9 discusses limitations honestly, and Section 10 concludes.

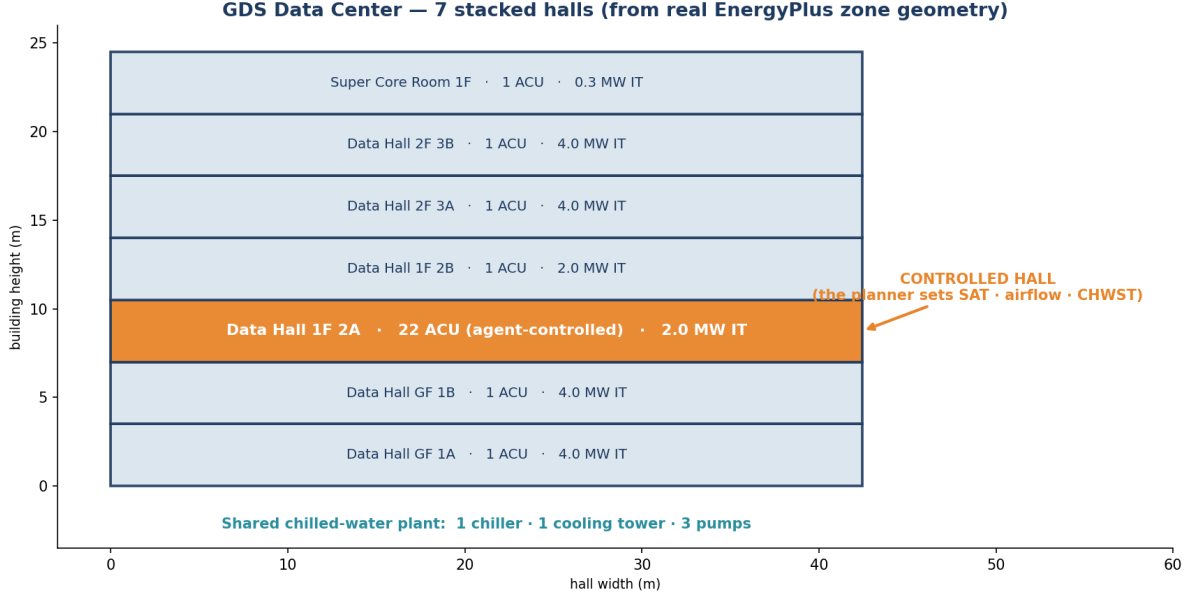


Figure 1: The modeled data center: seven stacked halls drawn to scale from the real EnergyPlus zone geometry. The orange hall (1F 2A) is the controlled hall — 22 agent-controlled CRAHs and ≈ 2.0 MW of IT load — served by a shared chilled-water plant. The planner sets only this hall’s supply-air temperature, airflow, and the chilled-water supply temperature.

2 System Overview

2.1 The physical plant

The controlled facility is a seven-hall, water-cooled tropical data center modeled to scale in EnergyPlus from its real zone geometry (Fig. 1). We control a single production hall, designated **1F 2A**, which contains 22 computer-room air handlers (CRAHs) serving ≈ 2.0 MW of IT load and is served by a *shared* chilled-water plant (chillers, cooling towers, and pumps) common to the building. The remaining halls are part of the simulation but are not actuated by the planner.

Within the controlled hall (Fig. 2), the racks are arranged in standard hot-aisle/cold-aisle rows. The planner exposes only three *global* setpoints, which the building’s control logic broadcasts to the individual actuators:

- **SAT** — CRAH supply-air temperature, range $20\text{--}26^\circ\text{C}$;
- **Airflow** — CRAH supply airflow, range $4.8\text{--}13.8$ kg/s per ACU;
- **CHWST** — chilled-water supply temperature, range $13\text{--}19^\circ\text{C}$.

These three numbers expand to $22 \text{ SAT} + 22 \text{ airflow} + 1 \text{ CHWST} = 45$ per-actuator commands (Section 3.2). The overriding constraint is a **hard cap of 26°C on every server inlet at every 15-minute step**.

2.2 The dual-loop architecture

The framework (Fig. 3) is organized as two nested loops.

The inner planning loop runs entirely in simulation. A *forecaster* predicts the coming week’s IT load and weather; a *beam search* proposes candidate setpoint triples; an *oracle* scores each candidate with a full-week EnergyPlus run; an *objective* converts each candidate’s key performance indicators (KPIs) into a scalar cost while a *safety gate* rejects any candidate that

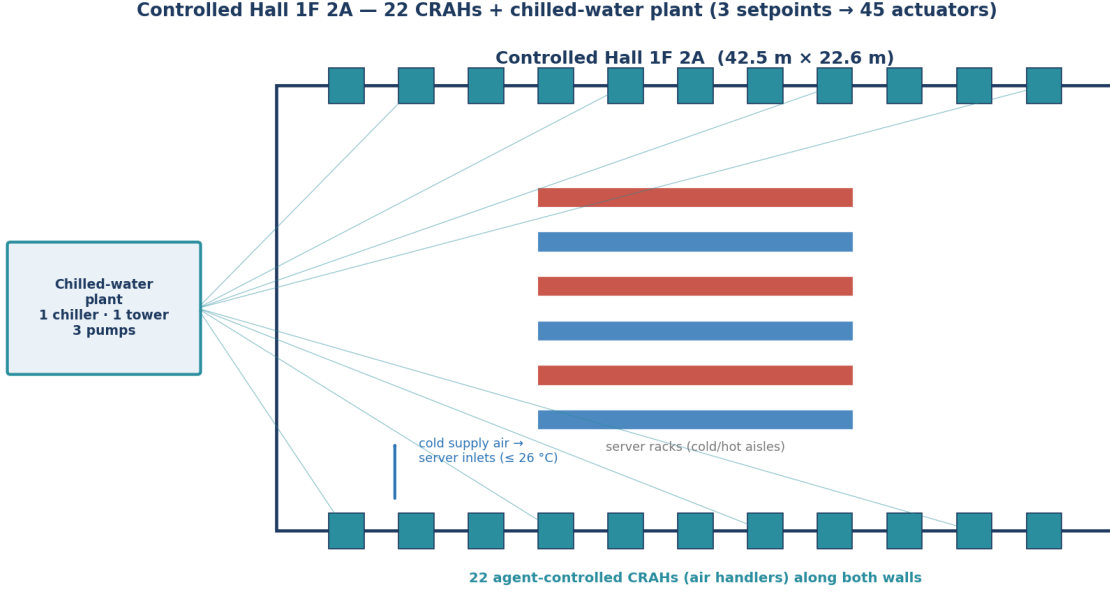


Figure 2: Plan view of the controlled hall 1F 2A: hot-aisle/cold-aisle rack rows, 22 CRAHs along the walls, and the shared chilled-water plant. The three global setpoints (SAT, airflow, CHWST) are broadcast to $22 + 22 + 1 = 45$ actuators that deliver cold supply air to the server inlets, which must remain at or below 26°C .

violates — or cannot be proven to robustly satisfy — the inlet cap. The result is a single recommended plan.

The *outer deployment loop* closes the system to operations. The recommended plan is *pre-validated* by independent nominal and worst-case replays; an expert *approves, edits, or rejects* it; on approval it is expanded to 45 commands and *deployed in shadow mode* (recorded, not physically actuated, pending field connection); the realized week is *measured*; and a *calibrator* updates the twin’s bias and uncertainty estimates, which re-enter the inner loop’s search margin and robustness ensemble on the next cycle. A slower *physics recalibration* corrects the plant model’s parameters themselves once a persistent energy bias accumulates.

A defining property of the design is that **the same first-principles engine is the oracle on both sides**: candidates are scored, finalists are stress-tested, and deployed weeks are re-evaluated by the identical EnergyPlus model, so the safety argument never leaves physics for a cheaper surrogate.

3 Physical and Mathematical Formulation

Throughout, a week is discretized into 15-minute control steps; with the default of four timesteps per hour, the step length is $\Delta t = 0.25$ h. We write t for the step index (after a warm-up discard of the first six samples), r for a server-rack (ITE) index, and z for a thermal zone. The hard inlet cap is $T_{\text{cap}} = 26.0^\circ\text{C}$. Table 2 (Appendix A) collects every constant.

3.1 The EnergyPlus oracle and KPI aggregation

Each candidate setpoint triple is evaluated by a complete week-long EnergyPlus 9.5 simulation, run inside a Docker container and coupled to the controller through a BCVTB socket. Because the dctwin/EnergyPlus configuration is a process-global singleton, the oracle parallelizes *across processes*, fanning candidates out over a `ProcessPoolExecutor`. Each evaluation is bounded by a per-candidate watchdog (300 s), a batch deadline, and a stall watchdog that abandons stragglers

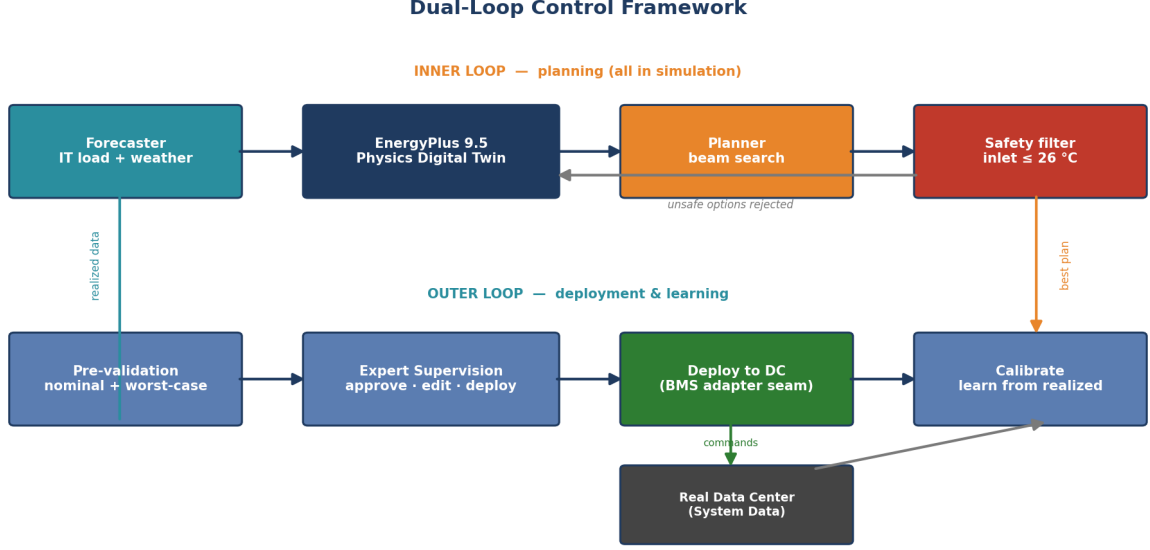


Figure 3: The dual-loop control framework. *Inner loop* (planning, all in simulation): forecaster → EnergyPlus physics twin → beam search → safety filter, with unsafe candidates rejected. *Outer loop* (deployment and learning): pre-validation → expert supervision → deploy → calibrate from realized data, which feeds the forecaster and search of the next cycle.

if nothing completes within $1.5\times$ the timeout — so a hung container costs minutes, not the whole plan.

From the per-step EnergyPlus output the oracle computes the controllable HVAC power. The energy objective is *hall-scoped* (scope `hall_controllable_v1`): it sums the 1F-2A ACU fan powers and the shared chiller/CHW-plant powers, rather than the cruder “facility total minus IT.” Writing $\mathcal{H}$ for the set of discovered controllable power channels (ACU fans $\cup$ chiller/CHW plant) and $P_{j,t}$ for channel j ’s power at step t ,

$$P_t^{\text{hvac}} = \begin{cases} \sum_{j \in \mathcal{H}} P_{j,t}, & \text{if controllable channels are discovered,} \\ P_t^{\text{total}} - P_t^{\text{IT}}, & \text{fallback (mock/legacy).} \end{cases} \quad (1)$$

The fallback (facility power minus IT power) is used only when the component channels are not present, e.g. in unit-test stand-ins. Per-step energy and the weekly total follow as

$$E_t = \frac{P_t^{\text{hvac}} \Delta t}{1000} [\text{kWh}], \quad E_{\text{hvac}} = \sum_t E_t [\text{kWh}], \quad (2)$$

where the factor $1/1000$ converts $\text{W}\cdot\text{h}$ to kWh. The mean Power Usage Effectiveness is computed facility-wide and only over steps with non-zero IT power,

$$\overline{\text{PUE}} = \frac{1}{|\{t : P_t^{\text{IT}} > 0\}|} \sum_{t: P_t^{\text{IT}} > 0} \frac{P_t^{\text{total}}}{P_t^{\text{IT}}}. \quad (3)$$

The safety-relevant KPIs are the peak inlet temperature, the count of hard violations, and a soft excess accumulator,

$$T_{\max}^{\text{inlet}} = \max_t \left(\max_r T_{r,t} \right), \quad (4)$$

$$N_{\text{viol}}^{\text{inlet}} = \sum_t \mathbb{1} \left[\max_r T_{r,t} > T_{\text{cap}} \right], \quad (5)$$

$$S^{\text{inlet}} = \sum_t \max \left(\max_r T_{r,t} - T_{\text{soft}}, 0 \right), \quad T_{\text{soft}} = T_{\text{cap}} - 1.0 = 25.0^\circ\text{C}, \quad (6)$$

where $T_{r,t}$ is the inlet dry-bulb temperature at rack r and step t . Two further soft accumulators penalize relative-humidity excursions outside $[30\%, 60\%]$ and zone-temperature deviation from the tropical band $32 \pm 1^\circ\text{C}$:

$$S^{\text{rh}} = \sum_t \sum_h \left[\max(\text{RH}_{\min} - \text{RH}_{h,t}, 0) + \max(\text{RH}_{h,t} - \text{RH}_{\max}, 0) \right], \quad (7)$$

$$S^{\text{zone}} = \sum_t \sum_z \max(|T_{z,t}^{\text{zone}} - T_{\text{zone}}^*| - b_{\text{zone}}, 0), \quad (8)$$

with $\text{RH}_{\min} = 30\%$, $\text{RH}_{\max} = 60\%$, $T_{\text{zone}}^* = 32^\circ\text{C}$, and half-band $b_{\text{zone}} = 1^\circ\text{C}$.

3.2 Control parametrization and the broadcast map

The decision variable is the triple (SAT, flow, CHWST) within the box defined above. EnergyPlus actuators are driven in a normalized $[-1, 1]$ coordinate; each physical value x with bounds (lb, ub) maps linearly as

$$\tilde{x} = \frac{2(x - \text{lb})}{\text{ub} - \text{lb}} - 1 \in [-1, 1]. \quad (9)$$

The three globals are then *broadcast* to the 45-dimensional action vector by repeating each global across its actuator group,

$$\underbrace{22}_{\text{SAT}} + \underbrace{22}_{\text{airflow}} + \underbrace{1}_{\text{CHWST}} = 45, \quad (10)$$

each entry normalized by Eq. (9) with its own bounds. Reducing the search to three shared setpoints is both physically reasonable (a single hall's CRAHs are operated in concert) and what makes a full-physics search tractable.

3.3 The objective function

Candidates are ranked by a single scalar cost that is dominated by energy and shaped by the soft penalties. When a time-of-use tariff or grid-carbon profile is supplied, the energy term is replaced by its hour-weighted cost,

$$C_{\text{wgt}} = \sum_t E_t \cdot \rho[(h_0 + \lfloor t \Delta t \rfloor) \bmod 24], \quad (11)$$

where ρ is the vector of 24 hourly rates (in $\$/\text{kWh}$ or kgCO_2/kWh) and h_0 is the hour-of-day of the first scored step. The scalar objective (lower is better) is

$$J = E^* + \lambda_T S^{\text{inlet}} + \lambda_{\text{rh}} S^{\text{rh}} + \lambda_z S^{\text{zone}}, \quad E^* = \begin{cases} C_{\text{wgt}}, & \text{tariff set,} \\ E_{\text{hvac}}, & \text{otherwise,} \end{cases} \quad (12)$$

with weights $\lambda_T = 1.0$ (per $^\circ\text{C}$ -step of inlet excess), $\lambda_{\text{rh}} = 0.2$, $\lambda_z = 0.1$. An infeasible candidate (next section) is assigned $J = +\infty$; so is any candidate whose computed cost is non-finite. Crucially, *feasibility never trades against price*: the soft penalties shape the search among *feasible* options, they cannot buy a violation. The headline operational metric is the energy reduction against the as-operated baseline,

$$\Delta E_{\%} = \frac{E_{\text{base}} - E_{\text{plan}}}{E_{\text{base}}} \times 100, \quad (13)$$

where E_{base} is the energy of the as-operated setpoints (the historical operating point, Section 6.3) evaluated once by the same oracle.

3.4 Forecasting the week ahead

Both the IT load and the weather for the coming week are forecast from history, since the planner runs a week before the controlled period.

IT load. The per-hall IT load L_t (kW) is converted to a CPU-loading fraction against nameplate capacity $C = \text{total_watts}/1000$ (kW),

$$u_t = \text{clip}\left(\frac{L_t}{C}, 0, 1\right). \quad (14)$$

A seasonal climatology is built by bucketing historical observations by weekday and time-of-day bin, and the forecast for step i is the empirical percentile of its bucket,

$$\hat{u}_i^{(q)} = \text{percentile}_q(\{u : \text{same (weekday, time-bin) bucket}\}), \quad q \in \{10, 50, 90\}, \quad (15)$$

giving p10/p50/p90 bands; the point forecast is the p50. The pooled diurnal shape is then re-leveled to the target period by a clipped ratio of local-to-global daily means,

$$\kappa = \text{clip}\left(\frac{\mu_{\text{local}}}{\mu_{\text{global}}}, 0.5, 1.5\right), \quad (16)$$

with μ_{local} the mean of whole-day means in a ± 10 -day window around the week start and μ_{global} the all-history mean. For this hall the IT load is nearly flat month-to-month, so in practice *weather* drives the week-to-week variation.

Weather. For the target week we compute historical-analog statistics of the outdoor dry-bulb temperature over all years, using rows whose month-day falls within the target week padded by ± 7 days,

$$\mu_c = \text{mean}(\{T_{\text{db}}\}), \quad \sigma_c = \text{pstdev}(\{T_{\text{db}}\}). \quad (17)$$

From these we synthesize EnergyPlus weather-file (EPW) variants by shifting the entire dry-bulb series,

$$\delta = k \sigma_c, \quad \{\text{nominal} : 0, \text{hot} : +\delta, \text{cool} : -\delta\}, \quad k = 1, \quad (18)$$

with the dew point clamped not to exceed the shifted dry-bulb. The $+1\sigma$ “hot” variant is added as an extra scenario in the robustness gate (Section 5), so the plan must hold the inlet cap even in an unusually hot week.

3.5 The recirculation fidelity model

The link between a CRAH’s supply-air temperature and a server’s *inlet* temperature is not the identity: in any real hall some fraction of hot exhaust air recirculates into the cold aisle. We model this with the standard mixing identity — the inlet is a blend of supply and return air,

$$T_{\text{inlet}} = r T_{\text{return}} + (1 - r) T_{\text{supply}} \implies r = \frac{T_{\text{inlet}} - T_{\text{supply}}}{T_{\text{return}} - T_{\text{supply}}}, \quad (19)$$

where $r \in [0, 1]$ is the recirculation fraction. Given measured rack temperatures the fraction is estimated robustly as a clipped median, discarding steps whose supply/return gap is too small to be informative ($< 1^\circ\text{C}$):

$$\hat{r} = \text{median}\left\{\text{clip}(r, 0, R_{\text{max}}) : |T_{\text{return}} - T_{\text{supply}}| \geq 1^\circ\text{C}\right\}, \quad R_{\text{max}} = 0.5. \quad (20)$$

Recirculation worsens when CRAH airflow undershoots the IT demand, so we add a flow-shortfall term to the calibrated baseline fraction r_0 ,

$$r_{\text{eff}} = \min\left(R_{\text{max}}, r_0 + k_r \max\left(0, 1 - \frac{\text{flow}}{\text{demand}}\right)\right), \quad k_r = 0.5, \quad (21)$$

and apply a *conservative-only* (safety-tightening) correction to the predicted inlet,

$$T'_{\text{inlet}} = T_{\text{inlet}} + \max\left(0, (r_{\text{eff}} - r_0) \max(0, T_{\text{zone}} - T_{\text{inlet}})\right). \quad (22)$$

When the airflow meets demand, $r_{\text{eff}} = r_0$ and the correction is exactly zero; the model can only raise the inlet, never lower it. Until calibrated against real rack sensors the correction defaults to a provable no-op, so no inlet margin is invented.

4 Optimization: Coarse-to-Fine Beam Search

The inner loop searches the three-dimensional setpoint box with a best-first beam search that refines a coarse grid. At level 0 it evaluates a regular grid,

$$\text{grid} = \text{linspace}(\text{lb}, \text{ub}, g)^{\otimes 3}, \quad g = 5 \Rightarrow 5^3 = 125 \text{ candidates}, \quad (23)$$

keeps the top- B (beam width $B = 5$) by score, and refines each survivor by exploring a neighborhood whose step is half the coarse spacing and halves again every level,

$$\Delta_d^{(0)} = \frac{\text{ub}_d - \text{lb}_d}{2(g-1)}, \quad \Delta_d^{(\ell+1)} = \frac{1}{2} \Delta_d^{(\ell)}, \quad d \in \{\text{SAT}, \text{flow}, \text{CHWST}\}. \quad (24)$$

Each candidate's neighbors are axis and pairwise-diagonal offsets clipped to the box; the beam is re-selected by a stable ascending sort on J (infeasible candidates, scored $+\infty$, sink to the bottom). Refinement stops early when the best score stops improving,

$$|J_{\text{best}}^{(\ell-1)} - J_{\text{best}}^{(\ell)}| < \epsilon \max(|J_{\text{best}}^{(\ell-1)}|, 1), \quad \epsilon = 10^{-3}. \quad (25)$$

With the default three levels this evaluates a few hundred full-week simulations per plan. An optional time-block extension splits the week into day [6, 18) and night blocks and warm-starts a coordinate-descent refinement (initial step a quarter of the range, halving per level) from the constant-setpoint optimum, allowing distinct day/night setpoints when they help.

Degenerate-signal guard. A physics model wired incorrectly can be *control-invariant* — every setpoint yields the same energy and inlet, so the search optimizes noise. The optimizer therefore flags a degenerate level when both the inlet and energy spreads collapse,

$$\left(\max_i T_{\text{max},i}^{\text{inlet}} - \min_i T_{\text{max},i}^{\text{inlet}}\right) < 0.1 \text{ } ^\circ\text{C} \quad \wedge \quad \frac{\max_i E_i - \min_i E_i}{\overline{E}} < 0.01. \quad (26)$$

The optimization is physically well-posed. A direct EnergyPlus sweep (Fig. 4) confirms that the three setpoints exert a real, well-separated physical gradient on energy while the inlet cap binds: across its range *airflow* moves hall cooling energy by 15.5%, *CHWST* by 3.4%, and *SAT* by $\approx 0.5\%$. The cheapest operating point (lowest airflow) is also the one that runs the inlet closest to the cap, which is exactly why a search with a hard constraint — not an unconstrained minimization — is required.

5 Safety: The Three-Net Robust Gate

Safety is enforced by three independent nets (Fig. 5), each strictly necessary, with model uncertainty hedged exactly once.

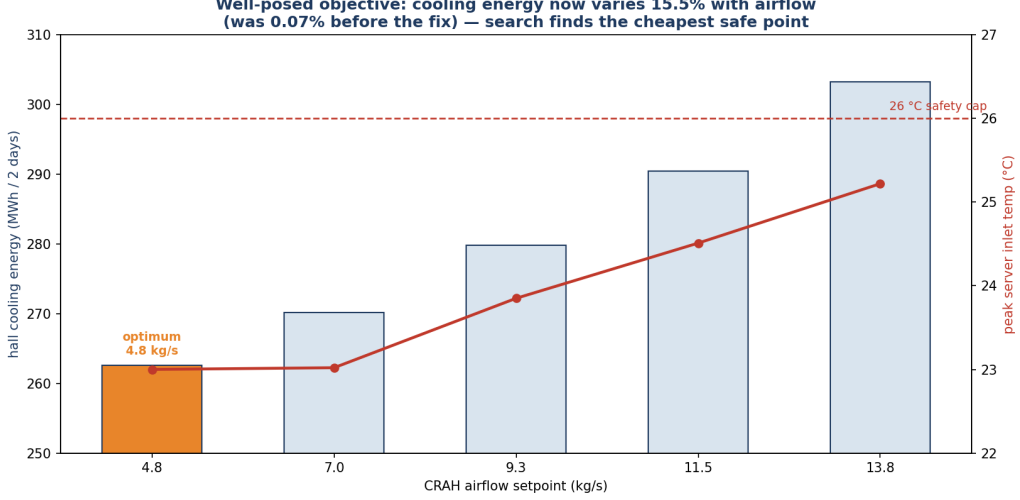


Figure 4: Well-posedness sweep (real EnergyPlus). Bars: hall cooling energy versus CRAH airflow; line: peak server inlet versus airflow, against the 26°C cap. Cooling energy varies 15.5% across the airflow range (3.4% for CHWST, $\approx 0.5\%$ for SAT) — a true physical gradient — while the cheapest (lowest-airflow) point is the hottest, so the constrained search is meaningful.

5.1 Net 1 — hard cap with a $k\sigma$ margin

A candidate is *feasible* only if it satisfies all of:

$$\text{feasible} \wedge (N_{\text{viol}}^{\text{inlet}} \leq 0) \wedge (m_{\text{fc}} = 0 \vee T_{\text{max}}^{\text{inlet}} + m_{\text{fc}} \leq T_{\text{cap}}), \quad (27)$$

i.e. *zero* steps above the cap, and the peak inlet at least a margin m_{fc} below it. The margin pre-tightens the cap by k_σ standard deviations of the twin’s inlet-prediction error,

$$m_{\text{fc}} = k_\sigma \sigma_{\text{inlet}}, \quad k_\sigma = 1.0, \quad \sigma_{\text{inlet}} = \begin{cases} \sigma_{\text{inlet}}^{\text{floor}}, & n_{\text{weeks}} > 0, \\ 1.0^\circ\text{C}, & n_{\text{weeks}} = 0 \text{ (cold start)}, \end{cases} \quad (28)$$

where σ^{floor} is the fading-floor calibration uncertainty (Section 6.1). At cold start the effective cap is $26 - 1.0 = 25.0^\circ\text{C}$; as weeks accumulate and the twin earns confidence, σ_{inlet} shrinks and more of the 26°C headroom becomes usable.

5.2 Net 2 — a data-centered robust ensemble

The few finalists from the search are re-simulated on an *ensemble* of perturbed plants, plus the $+1\sigma$ hot-weather week of Eq. (18). The ensemble is centered on the *believed* plant (the data-driven plant parameters fitted by recalibration, Section 6.2; a fixed default until then), and its width is set by the posterior uncertainty. Scenario i scales each plant perturbation factor by

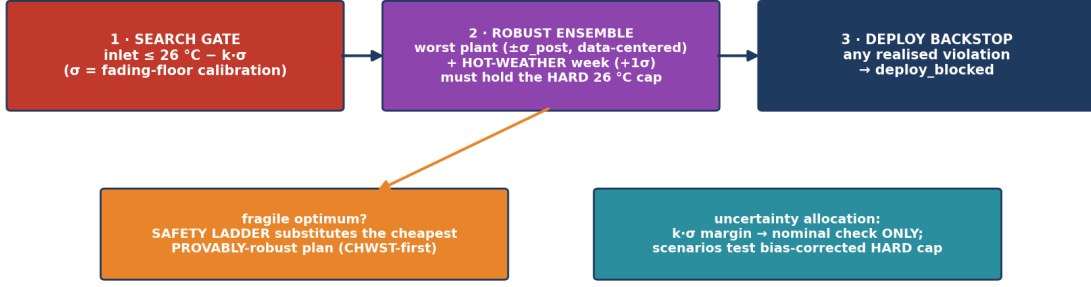
$$m_i = (1 - s) + 2s \cdot \frac{i}{n - 1}, \quad i = 0, \dots, n - 1, \quad (29)$$

where the half-width s adapts to measured confidence,

$$s = \min\left(s_{\text{base}}, \max\left(s_{\text{min}}, s_{\text{base}} \cdot \frac{\sigma_{\text{post}, \text{inlet}}}{\sigma_{\text{ref}}}\right)\right), \quad s_{\text{base}} = 0.10, \quad s_{\text{min}} = 0.02, \quad \sigma_{\text{ref}} = 1.0, \quad (30)$$

so the spread never falls below a $\pm 2\%$ physical-drift floor and never exceeds the $\pm 10\%$ cold-start bracket. Inside these scenarios the $k\sigma$ margin is switched *off* ($m_{\text{fc}} = 0$): the uncertainty is spent once, in the ensemble width, not double-counted against a margin as well. Across the scenario

Three independent safety nets — uncertainty hedged exactly once



measured outcome: 6 deployed weeks · worst inlet 25.60 °C · 0 violations

Figure 5: The three-net safety architecture. Net 1: the hard 26 °C cap pre-tightened by a $k\sigma$ margin in the search. Net 2: a data-centered worst-case ensemble (plant $\pm\sigma_{\text{post}}$ plus a $+1\sigma$ hot-weather week) that must hold the *hard* cap, with uncertainty spent exactly once and a safety ladder substituting the cheapest provably-robust plan for a fragile optimum. Net 3: a zero-tolerance deploy backstop. Measured outcome over six deployed weeks: worst inlet 25.60 °C, zero violations.

KPIs the gate summarizes the energy tail with the conditional value-at-risk and reports robust bands by quantiles,

$$\text{CVaR}_\alpha(\{v_i\}) = \frac{1}{k} \sum_{j=1}^k v_{(j)}^\downarrow, \quad k = \max(1, \lceil (1 - \alpha)N \rceil), \quad \alpha = 0.8, \quad (31)$$

with $v^\downarrow$ the energies sorted worst-first. A finalist is **robustly feasible** iff a majority of its scenarios complete and *every* completed scenario is feasible by Eq. (27) (with scenario KPIs bias-corrected and tested against the hard cap):

$$|\mathcal{K}_i| \geq \left\lceil \frac{n_{\text{req}}}{2} \right\rceil \quad \wedge \quad \forall k \in \mathcal{K}_i : \text{is_feasible}(k). \quad (32)$$

The winner is the robustly feasible finalist of least energy CVaR.

The safety ladder. If the energy optimum is *fragile* — cheap on the nominal plant but not robustly feasible — the gate does not simply block. It walks a *safety ladder* from the chosen point toward the guaranteed-safe corner (SAT, flow, CHWST) = (20.0, 13.8, 13.0), spending the *cheapest* axis first. Because CHWST moves energy only $\approx 3\%$ while airflow moves it $\approx 15\%$ (Section 4), the ladder lowers chilled-water temperature before it raises airflow, substituting the least-costly provably-robust plan rather than returning nothing actionable.

5.3 Net 3 — the deploy backstop

After a week is realized, any single inlet violation marks the plan `deploy_blocked` rather than `deployed`. This zero-tolerance backstop closes the loop: a plan that slipped through prediction is never recorded as a clean deployment.

6 The Outer Loop: Learning from Reality

The twin improves itself on two timescales: a fast *output* calibration after every deployed week, and a slow *parameter* recalibration once a bias persists.

6.1 Output calibration: bias and two uncertainties

After each deployed week the paired (predicted p , realized r) values of three KPIs — weekly HVAC energy, mean PUE, and peak inlet — update the calibration state. Residuals are winsorized to a per-KPI clip c to keep one anomalous week from poisoning the estimate, and the bias is their mean,

$$e_w = \text{clip}(r - p, -c, +c), \quad b = \frac{1}{n} \sum_{w=1}^n e_w, \quad (33)$$

with n the number of paired weeks. Two uncertainties are maintained from the same residuals. The *fading-floor* σ , which backs the $k\sigma$ search margin of Eq. (28), never understates early because it is floored by a decaying prior,

$$\sigma^{\text{floor}} = \max\left(s, \frac{\sigma_{\text{prior}}}{n}\right), \quad s = \sqrt{\frac{1}{n} \sum_{w=1}^n (e_w - b)^2}, \quad (34)$$

so that at $n = 1$ (where the sample std $s = 0$) the uncertainty equals the prior rather than a dangerous zero, and the floor σ_{prior}/n decays as evidence accumulates. The *empirical-Bayes posterior* σ , which sizes the robust ensemble of Eq. (30), blends the sample variance with the prior as if the prior were one pseudo-week [13],

$$\sigma_{\text{post}} = \sqrt{\frac{n s^2 + \sigma_{\text{prior}}^2}{n + 1}}. \quad (35)$$

It tends to σ_{prior} at $n = 0$ and to s as $n \rightarrow \infty$. The inlet priors are $\sigma_{\text{prior}} = 1.0^\circ\text{C}$ for the fading floor. Finally, the learned bias is applied to future predictions, and — because a bias added to a peak cannot reconstruct per-step counts — if a bias-corrected peak inlet would exceed the cap, the violation count is forced to at least one so the feasibility gate rejects it:

$$x'_{\text{field}} = x_{\text{field}} + b_{\text{field}}; \quad \text{if } T_{\text{max}}^{\text{inlet}} + b > 26.0 : N_{\text{viol}}^{\text{inlet}} \leftarrow \max(N_{\text{viol}}^{\text{inlet}}, 1). \quad (36)$$

6.2 Physics recalibration: learning the plant's parameters

Output calibration corrects the twin's *predictions*; physics recalibration corrects the twin's *model*. Once at least four weeks show a persistent energy bias, the system fits a bounded fan-efficiency correction. The per-week realized/predicted energy ratio is winsorized and averaged to a persistent bias fraction,

$$\varrho_i = \text{clip}\left(\frac{r_i}{p_i}, 0.5, 2.0\right), \quad b = \frac{1}{m} \sum_{i=1}^m \varrho_i - 1, \quad (37)$$

and converted to a multiplicative correction on the EnergyPlus fan total efficiency,

$$\phi = \text{clip}\left(\frac{1}{1+b}, 0.85, 1.15\right), \quad (38)$$

clamped never to propose more than a $\pm 15\%$ efficiency change, and emitted only when $m \geq 4$ and $|b| \geq 0.01$. More realized energy than predicted ($b > 0$) yields $\phi < 1$ — the fans were modeled as too efficient. The fitted factor is persisted to the plant-calibration file and *becomes the center of the robust ensemble* (Section 5): the twin's plant parameters now learn, not just its outputs.

Closing the loop to reality — every seam built, shadow-safe today

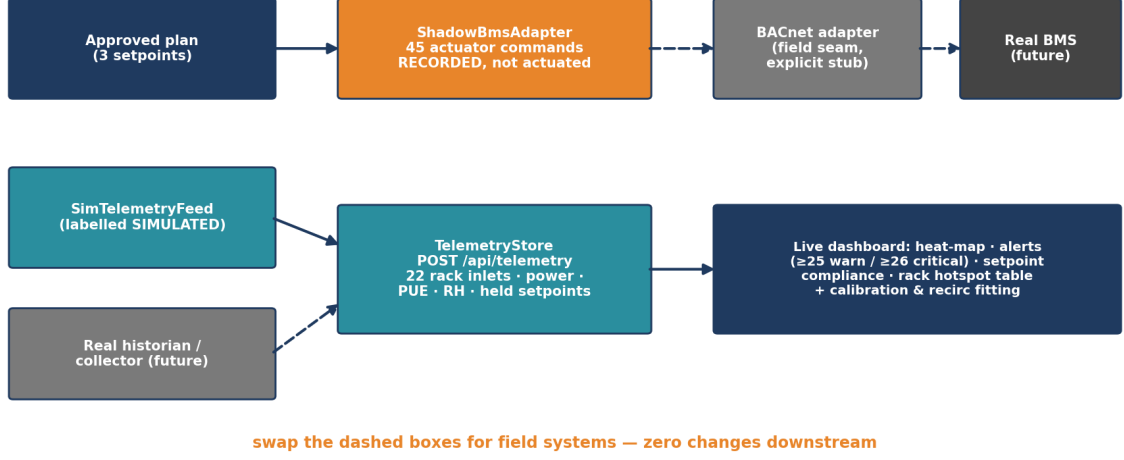


Figure 6: Closing the loop to reality. An approved plan becomes 45 actuator commands written by a shadow BMS adapter (recorded, not actuated); a telemetry store accepts any historian push (POST /api/telemetry) and feeds the live dashboard, calibration, and recirculation fitting. The dashed boxes — a real historian collector and a BACnet field adapter — are the only unbuilt seams; everything downstream is unchanged when they are connected.

6.3 The as-operated baseline

Savings are reported against the *as-operated* baseline — the operator’s own historical setpoints — not against an artificial worst case. Baseline SAT and CHWST are the pooled medians of the corresponding telemetry columns, and baseline airflow is derived from the median fan speed scaled to design flow,

$$\text{SAT}_{\text{base}} = \text{clip}(\widetilde{\text{SAT}}), \quad \text{CHWST}_{\text{base}} = \text{clip}(\widetilde{\text{CHWST}}), \quad \text{flow}_{\text{base}} = \text{clip}\left(\frac{\widetilde{\text{fan_speed}}}{\text{fan_speed}_{\text{max}}} \dot{m}_{\text{design}}\right), \quad (39)$$

where $\widetilde{(\cdot)}$ denotes the pooled median over matching telemetry columns and a missing signal falls back to the axis mid-range. The baseline is evaluated once by the same oracle and supplies E_{base} in Eq. (13).

7 Web Application and Operator Workflow

The framework is delivered as a single-origin web application (a FastAPI backend serving a React front end) so that an operator drives the loop without touching code. The backend exposes 21 endpoints, including a server-sent-events stream for live search progress and a telemetry push endpoint; the front end comprises seven pages. We highlight three.

The **Dashboard** (Fig. 7) shows the active plan’s setpoints and its predicted KPIs against the baseline — weekly energy, PUE, peak inlet, violation count, and percentage reduction — so the value and the safety margin of a recommendation are visible at a glance. The **Live Telemetry** page (Fig. 8) renders a 22-rack inlet heat-map, an alert banner (warn at $\geq 25^\circ\text{C}$, critical at $\geq 26^\circ\text{C}$), a ranked hotspot table, hall KPI tiles, and a commanded-versus-held setpoint-compliance panel that flags any axis drifting more than 0.5 from the deployed command; a prominent SIMULATED FEED badge marks the data source honestly. The **Digital Twin (3D)** page (Fig. 9) colors the hall’s rack rows live from telemetry, with a heads-up display of the active plan’s setpoints and KPIs, giving immediate spatial hotspot visibility.

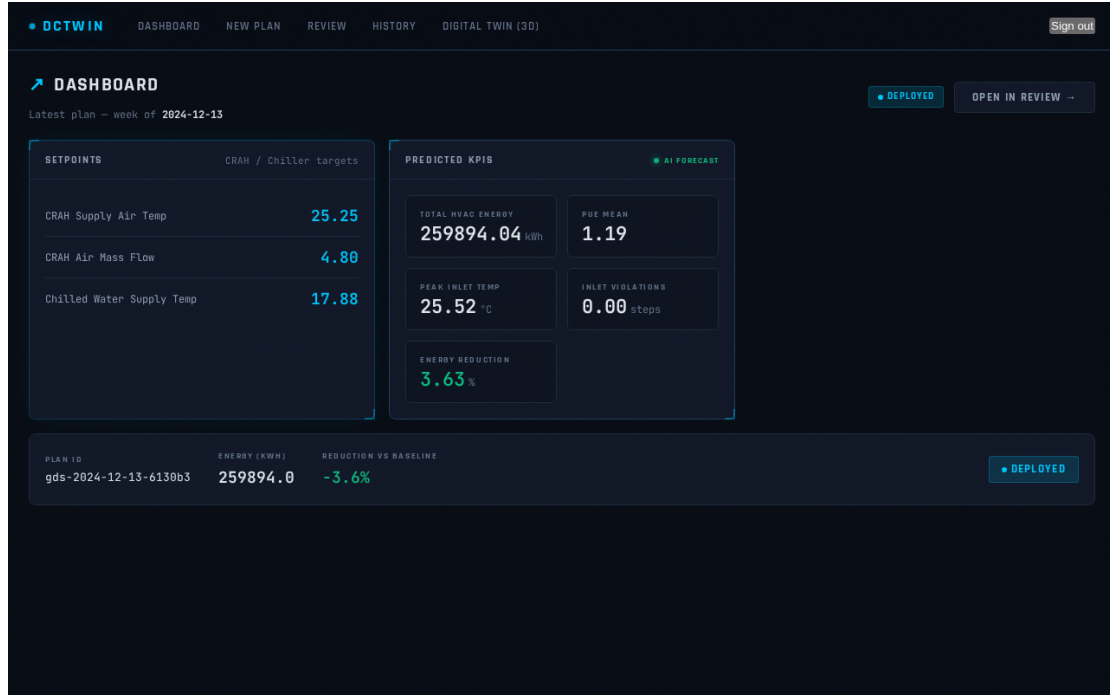


Figure 7: Dashboard (webapp screenshot). The active plan’s three setpoints (left) and its predicted KPIs against the as-operated baseline: weekly HVAC energy, PUE, peak inlet, and violations, with the energy reduction. For the displayed week the plan predicts a peak inlet of 25.52°C (below the 26°C cap), zero violations, and a 3.63% energy reduction.

The operator’s weekly cycle (Fig. 10) is a single turn of the dual loop: *monitor* the live hall, *plan* the coming week (reviewing the planning-context decks of forecast load and weather), *review* the recommendation’s KPIs and inlet trajectory against the 26°C line, *approve* as an expert (a `blocked_unsafe` plan cannot be force-approved — the gate refusing is the system working), *deploy* in shadow mode (45 commands recorded), and *learn* as the realized week tightens σ and unlocks more savings next time.

8 Experimental Results

We ran six consecutive weekly cycles — plan, gate, expert-approve, (shadow) deploy, measure, and learn — covering the weeks beginning 2024-11-08 through 2024-12-13. All numbers below are read from the live system’s run database and calibration history; deployment is in shadow mode, so the “realized” week is the same EnergyPlus model run on a perturbed (believed) plant standing in for the field until a real telemetry collector and BACnet adapter are connected (Section 9). Table 1 reports the per-week record.

The twin predicts reality (Fig. 11). The weekly-energy prediction error fell from 1.33% in week 1 to 0.10% by week 6, as the output calibration of Section 6.1 learned the twin’s residual bias. The safety-critical variable was predicted even better: the peak-inlet prediction error stayed within 0.25°C every week, with the calibrator settling on a small positive inlet bias ($\approx 0.18^{\circ}\text{C}$: the twin slightly *under*-predicts the hottest inlet, which the bias correction then adds back — a conservative direction for safety).

Savings released by earned confidence (Fig. 12). The energy saving against the as-operated baseline ramped from 0% in the cold-start week — where the gate, lacking calibration history, allowed essentially only the baseline — to 3.6% once measured confidence had grown.

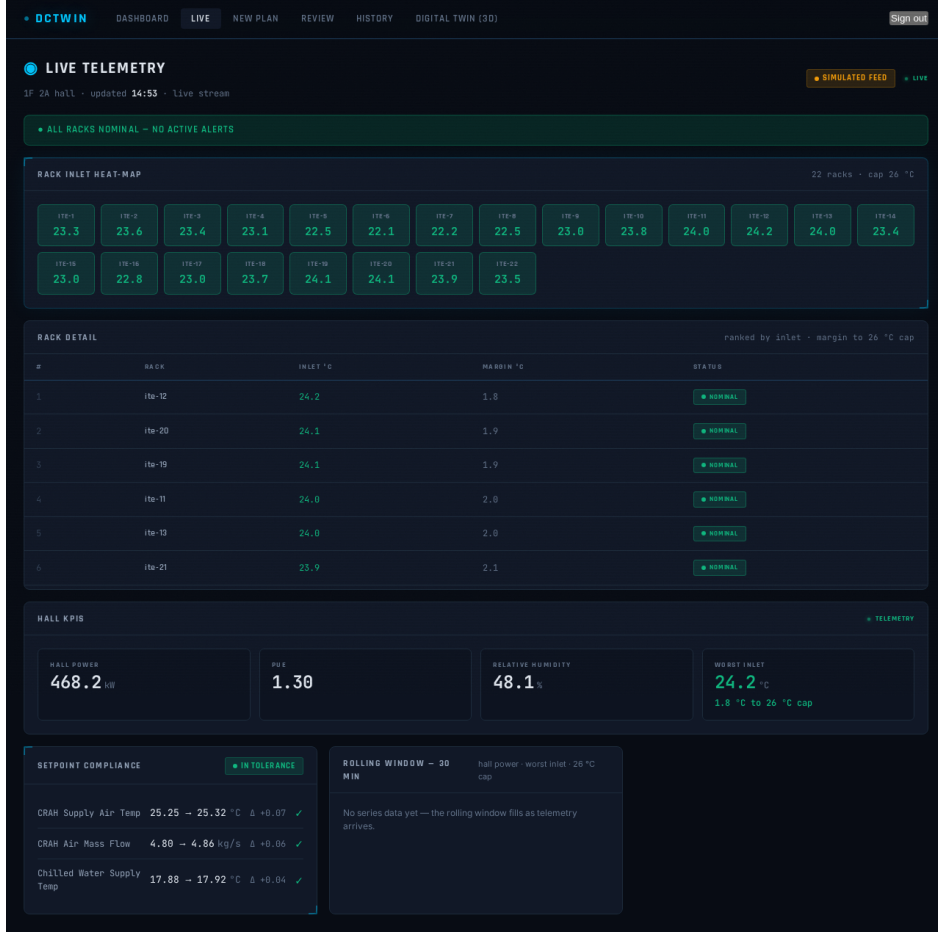


Figure 8: Live Telemetry (webapp screenshot). A 22-rack inlet heat-map (all green, $< 26^\circ\text{C}$), the SIMULATED FEED badge, a per-rack hotspot detail table, hall KPI tiles (PUE, peak inlet, etc.), and the commanded-versus-held setpoint-compliance panel. Alerts warn at $\geq 25^\circ\text{C}$ and turn critical at $\geq 26^\circ\text{C}$.

Table 1: Six consecutive deployed weeks. Predicted and realized weekly hall cooling energy, absolute prediction error, energy saving versus the as-operated baseline, realized peak server inlet, and hard-cap violations. Energy and inlet are read from the calibration history; savings from the run database.

Week (start)	Pred. energy (kWh)	Real. energy (kWh)	Err (%)	Saving (%)	Peak inlet ($^\circ\text{C}$)	Viol.
2024-11-08	273 565	269 979	1.33	0.00	23.37	0
2024-11-15	261 806	259 583	0.86	3.20	24.64	0
2024-11-22	260 480	259 929	0.21	3.23	24.68	0
2024-11-29	256 880	257 368	0.19	3.60	25.22	0
2024-12-06	258 531	258 739	0.08	3.64	25.57	0
2024-12-13	259 337	259 603	0.10	3.63	25.60	0

The ramp is not a tuning artifact: it is the direct consequence of the adaptive ensemble half-width of Eq. (30) narrowing as the posterior inlet uncertainty fell (from the 1.0°C prior toward the measured $\sigma_{\text{post}} \approx 0.36^\circ\text{C}$), which let the search safely approach the cap and spend the cheap energy axes. Savings rose *only as fast as* the system could prove them safe.

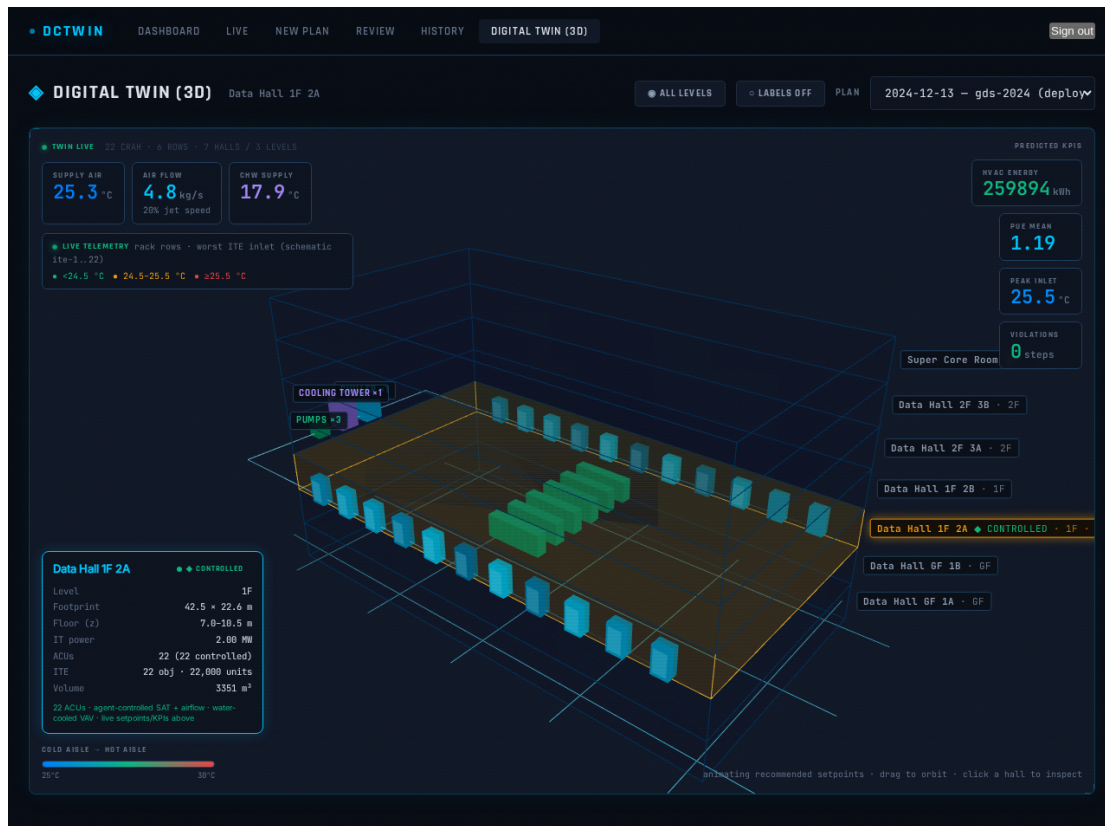


Figure 9: Digital Twin 3D (webapp screenshot). The controlled hall’s rack rows colored from live telemetry, with a heads-up display of the active plan’s setpoints (SAT, airflow, CHWST) and KPIs. Spatial coloring makes any emerging hotspot immediately visible.

The operator's week — one cycle of the dual loop

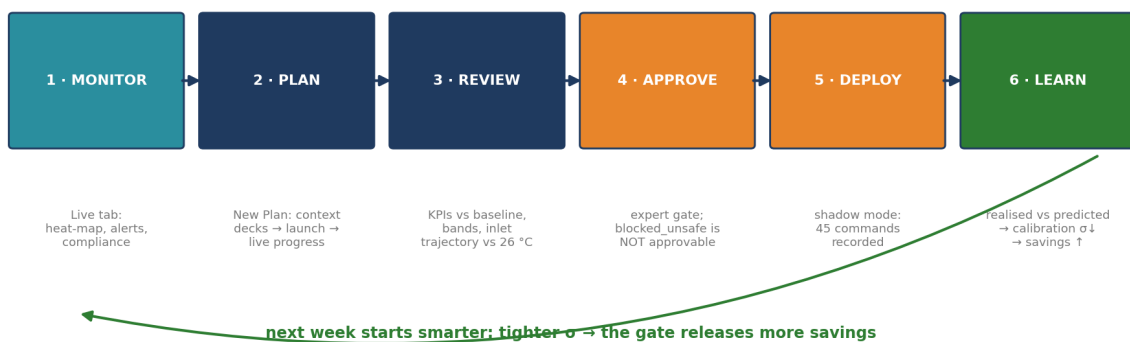


Figure 10: The operator’s week — one cycle of the dual loop: monitor → plan → review → approve → shadow deploy → learn. Each cycle measures the realized week, tightens the calibration uncertainty, and lets the safety gate release a little more saving.

Zero violations while saving (Fig. 13). Across all six deployed weeks there were *no* inlet violations. The realized peak inlet rose toward the cap as savings grew — from 23.37 °C to 25.60 °C — but never reached it, leaving a worst-case margin of 0.40 °C. This is the intended behaviour: the gate spends thermal margin only when calibrated confidence justifies it.

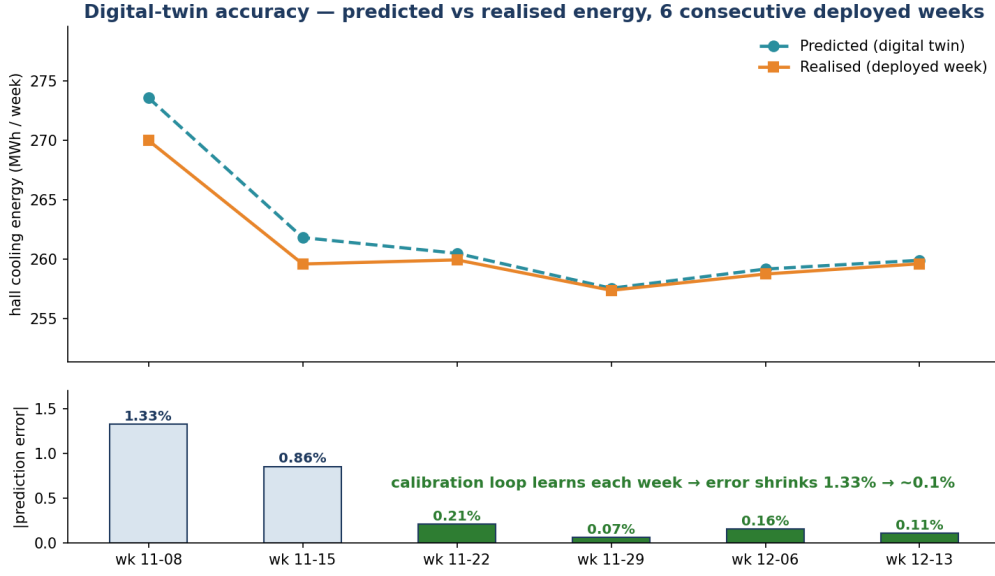


Figure 11: Digital-twin accuracy. Top: predicted versus realized weekly hall cooling energy over the six deployed weeks. Bottom: the absolute per-week prediction error, falling from 1.33% to $\approx 0.1\%$ as the residual-calibration loop learns.

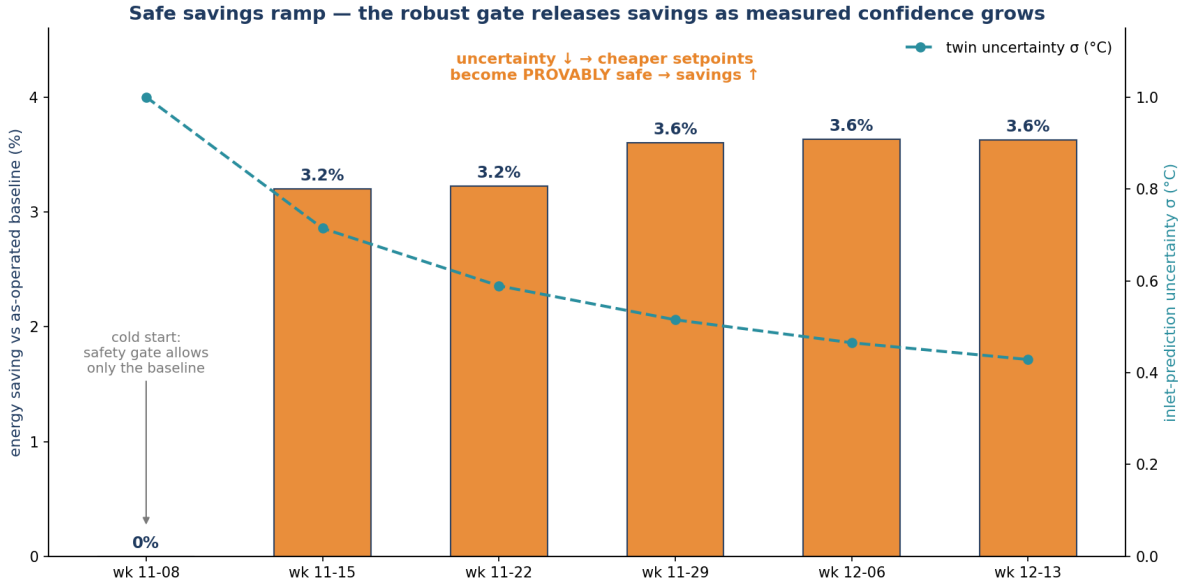


Figure 12: Safe savings ramp. Bars: per-week energy saving versus the as-operated baseline (0 $\rightarrow$ 3.6%). Line: the twin’s inlet-prediction uncertainty σ , falling from the 1.0 $^{\circ}\text{C}$ prior toward its measured value as weeks accumulate. The robust gate releases savings only as the proven-safe margin grows.

Calibration state. After seven paired weeks the live calibration state was a weekly-energy bias of $\approx 515 \text{ kWh}$ ($\approx 0.2\%$ of weekly energy), an inlet bias of 0.18°C , a fading-floor inlet σ of 0.14°C , and a posterior ensemble width σ_{post} of 0.36°C — the quantities that, through Eqs. (28) and (30), govern how aggressively the next plan may approach the cap. The whole framework is exercised by an automated test suite (422 backend and 104 front-end tests, plus nine Docker-gated EnergyPlus integration tests), all passing.

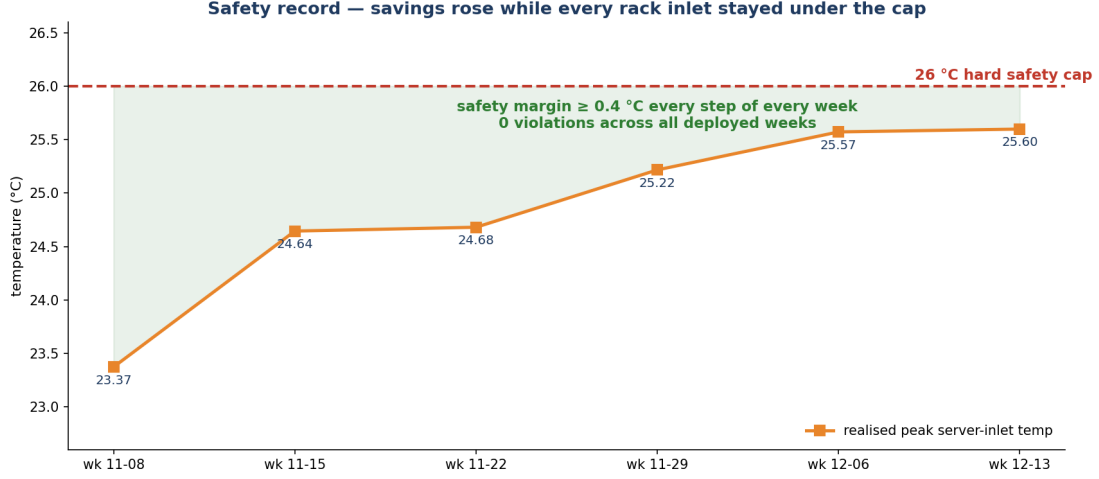


Figure 13: Safety record. Realized peak server inlet per week, rising toward — but never past — the 26 °C hard cap as savings grow; the shaded band is the realized safety margin. Worst realized inlet 25.60 °C; zero violations across all deployed weeks.

9 Discussion and Limitations

The results above are deliberately reported as *measured* rather than promised, and the same honesty applies to what is *not* yet field-proven. The single most important qualifier is that the system runs in **shadow mode**: an approved plan is expanded to the exact 45 per-actuator commands a field adapter would write, but those commands are *recorded*, *not actuated*. The BACnet/Modbus field adapter is an explicit, one-class seam, and the telemetry historian is a push endpoint awaiting a real collector (Fig. 6). Consequently the “realized” week is a perturbed-plant EnergyPlus simulation standing in for the building: the learning loop is *structurally* proven end-to-end, not yet *field*-proven. Three further model caveats follow from this. The recirculation correction of Eq. (22) is a provable no-op until calibrated against real rack-inlet sensors, so the inlet margin may be optimistic for halls with imperfect containment. The historical-analog weather σ of Eq. (17) understates true synoptic extremes; the $+1\sigma$ hot scenario only partially compensates. And because the IT load is nearly flat in this hall, the week-to-week story is dominated by weather, so absolute setpoints should be read as directional until recirculation is field-calibrated.

We also deliberately scoped out humidity as a controlled variable (it is monitored and penalized, but no dehumidification actuator is modeled), intra-week automatic re-planning (the cadence is weekly; urgent cases are handled by cancel-and-re-plan with the live alerts as a safety net), and end-to-end browser tests in continuous integration. None of these changes the framework; each is additive.

The path to a field pilot is correspondingly concrete and “config-shaped”: point a real collector at the telemetry endpoint, implement the BACnet adapter behind the existing seam, run in recommend-only mode for four to eight weeks comparing predictions to real measurements, calibrate recirculation from the first real rack-inlet data, and only then enable actuation hall-by-hall. The robustness machinery is designed to improve automatically as real weeks accumulate: σ_{post} tightens, the recalibrator re-centers the ensemble on the believed plant, and the objective can be switched from kWh to dollars or CO₂ simply by supplying the site’s tariff or grid-carbon profile to Eq. (11).

10 Conclusion

We presented a digital-twin dual-loop control framework for data-center cooling that keeps a first-principles EnergyPlus model in the loop as the oracle on both the planning and the deployment side, and that treats the server-inlet temperature cap as a hard constraint defended by three independent safety nets. The inner loop searches three global setpoints broadcast to 45 actuators, scoring each candidate by a full-week physics simulation; the outer loop pre-validates, deploys (in shadow), measures, and learns per-KPI bias and two uncertainties — a fading-floor σ that backs the search margin and an empirical-Bayes posterior that sizes a CVaR-based robust ensemble — while a slower physics recalibration corrects the plant parameters themselves. Over six consecutive deployed weeks the twin’s weekly-energy prediction error fell from 1.33% to 0.10%, savings ramped from 0% to 3.6% released strictly by earned confidence, and there were zero safety violations (worst inlet 25.60 °C). The remaining work to reach a live building is a field collector and a single adapter class behind seams that already exist — the rest of the loop is implemented, tested, and measured today.

Reproducibility

The framework is implemented in Python (planner and FastAPI backend) and TypeScript/React (front end), with EnergyPlus 9.5 executed in Docker and coupled through BCVTB. Plans, calibration state, and run artifacts are schema-versioned JSON; the per-week numbers in Table 1 and Figs. 11–13 are produced directly from the live run database and calibration history. Development followed a test-first discipline (tests are never weakened to pass), and an automated driver reproduces the build, smoke test, screenshot, and plan-execution steps.

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A Notation and Constants

Table 2: Key symbols, constants, and their values as used in the implementation.

Symbol / quantity	Meaning	Value / range
T_{cap}	hard inlet-temperature cap	$26.0\text{ }^{\circ}\text{C}$
SAT	CRAH supply-air temperature	$20\text{--}26\text{ }^{\circ}\text{C}$
flow	CRAH supply airflow (per ACU)	$4.8\text{--}13.8\text{ kg/s}$
CHWST	chilled-water supply temperature	$13\text{--}19\text{ }^{\circ}\text{C}$
Δt	control timestep	0.25 h (4/h)
T_{soft}	inlet soft-penalty threshold	$25.0\text{ }^{\circ}\text{C}$
$\lambda_T, \lambda_{\text{rh}}, \lambda_z$	objective penalty weights	1.0, 0.2, 0.1
$\text{RH}_{\text{min}}, \text{RH}_{\text{max}}$	RH band	30%, 60%
$T_{\text{zone}}^* \pm b_{\text{zone}}$	zone-temperature band	$32 \pm 1\text{ }^{\circ}\text{C}$
g, B, L	beam-search grid / width / levels	5, 5, 3
ϵ	early-stop tolerance	10^{-3}
k_{σ}	search-margin multiplier	1.0
α	CVaR confidence level	0.8
$s_{\text{base}}, s_{\text{min}}$	ensemble half-width / floor	0.10, 0.02
σ_{prior} (inlet)	inlet uncertainty prior	$1.0\text{ }^{\circ}\text{C}$
ϕ clip	fan-efficiency correction bound	[0.85, 1.15]
R_{max}, k_r	recirculation cap / shortfall gain	0.5, 0.5
oracle watchdog	per-candidate timeout	300 s